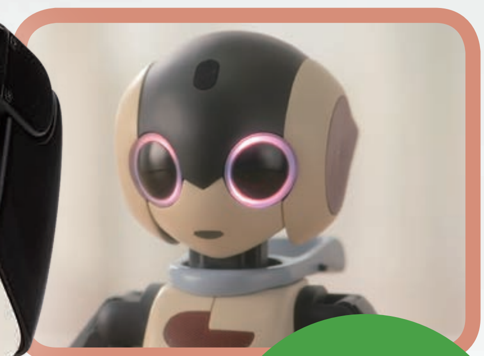
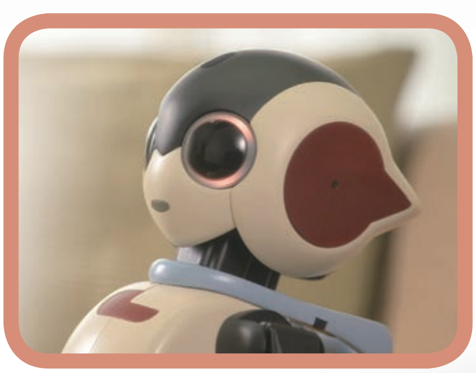
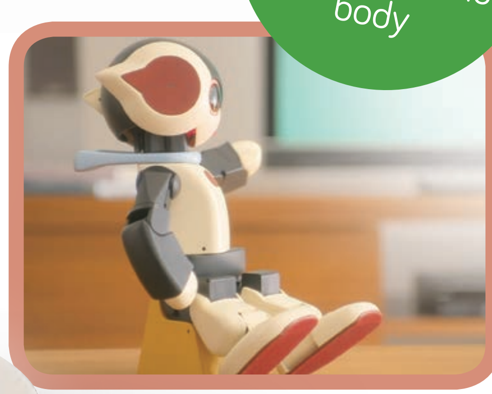
Build your own Robi

Pack 15







Stages 55-58: Work on Robi's head, neck and body

DeAGOSTINI MODEL SPACE

Build your own Robi

CONTENTS

DeAGOSTINI MODEL SPACE™

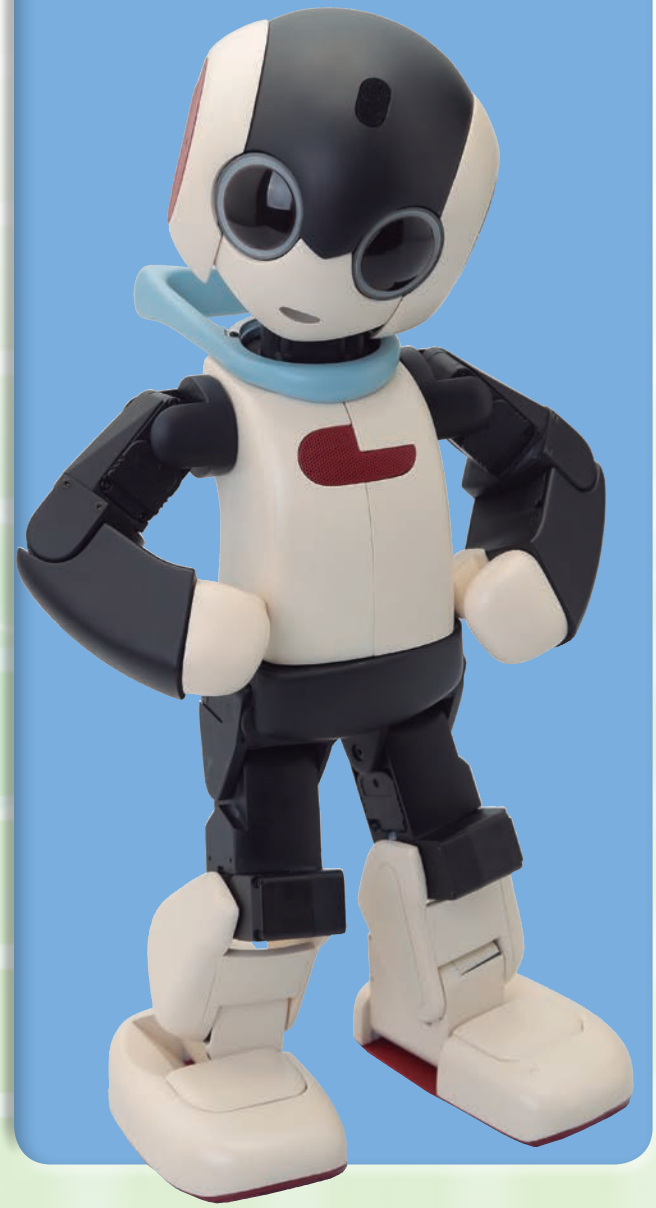
www.model-space.com

Published in the UK by De Agostini UK Ltd, Battersea Studios 2, 82 Silverthorne Road, London SW8 3HE

Published in the USA by De Agostini Publishing USA, Inc., 121 E. Calhoun Street Woodstock, IL 60098

Packaged by Continuo Creative, 39-41 North Road, London N7 9DP

Printed in EU

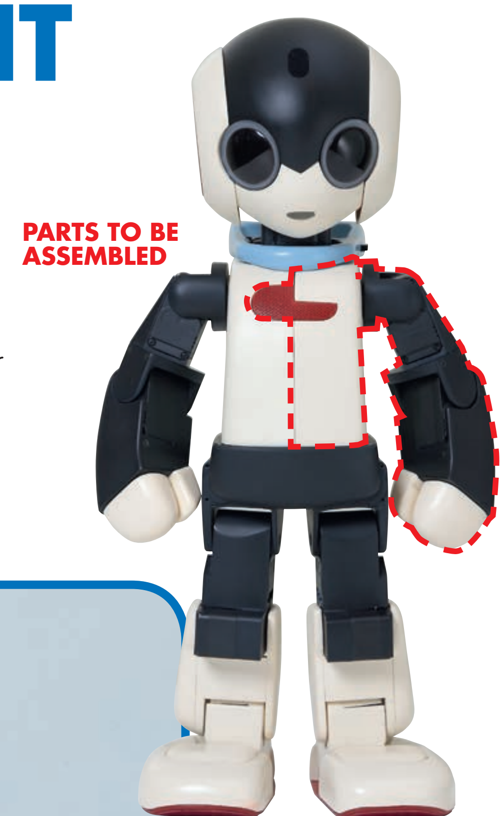


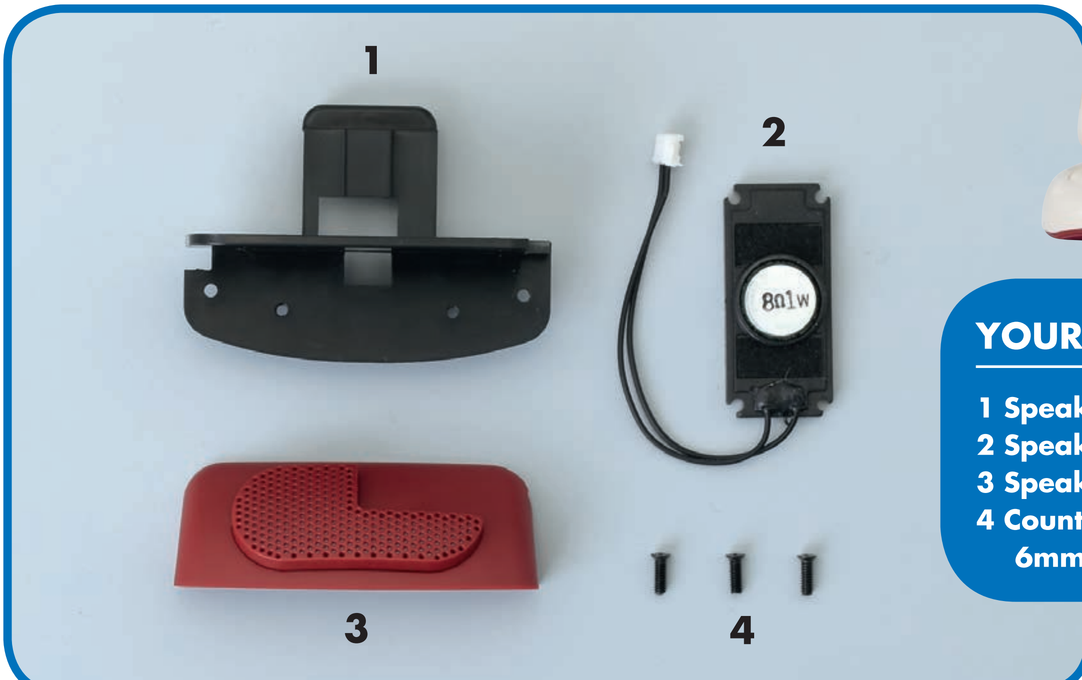
ASSEMBLY GUIDE

STAGE 55: MOUNT ROBI'S SPEAKER

In this session, you will fit the speaker to its cover and support, then fit these into the left side of Robi's previously assembled body panels.

This stage sees you work with one of Robi's more delicate components: the speaker from which Robi's voice will emanate on completion. The speaker's membrane is very delicate, so care must be taken when handling it – see the box at the foot of this page for more on this. Other than that, the assembly work is straightforward, although you must be careful not to trap any of the cables when securing the speaker in Steps 7-9.



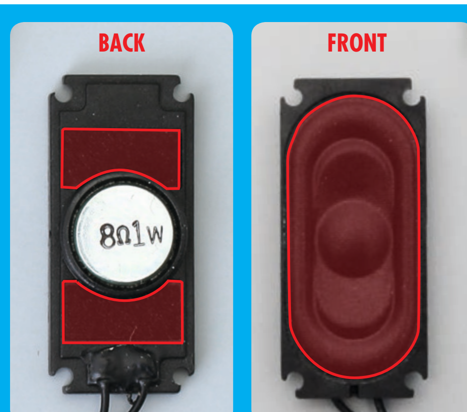


YOUR PARTS

1 Speaker support
2 Speaker
3 Speaker cover
4 Countersunk screws M2 x 6mm x 3 (one is a spare)

SAVED PARTS

Left arm and body panel (Stage 44)



DO NOT TOUCH THE AREAS HIGHLIGHTED IN RED!

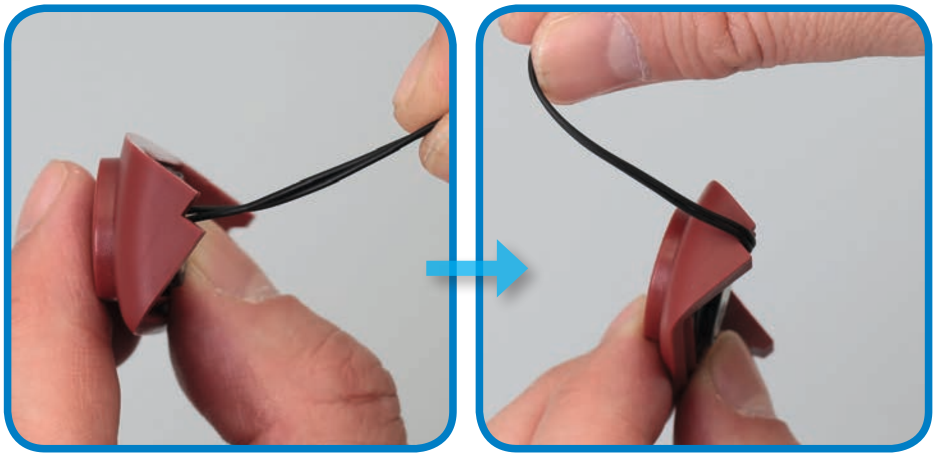
ATTENTION!

The speaker contains an electrical coil, which drives a very thin membrane that vibrates to project the sound of Robi's voice. It is very important that this be kept intact and does not get creased or torn in any way, as that will affect the sound! To ensure this, hold the speaker only from the sides and do not touch the areas marked in red.

MOUNTING THE SPEAKER

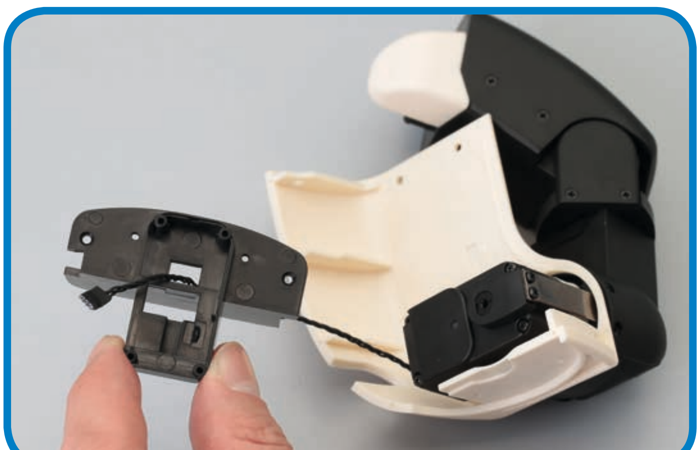
1 Line up the speaker with its front cover, orienting the parts as shown above. Remember not to touch the speaker in the areas marked in red on the previous page.

2 Fit the speaker to the cover. It should sit neatly within the recess.



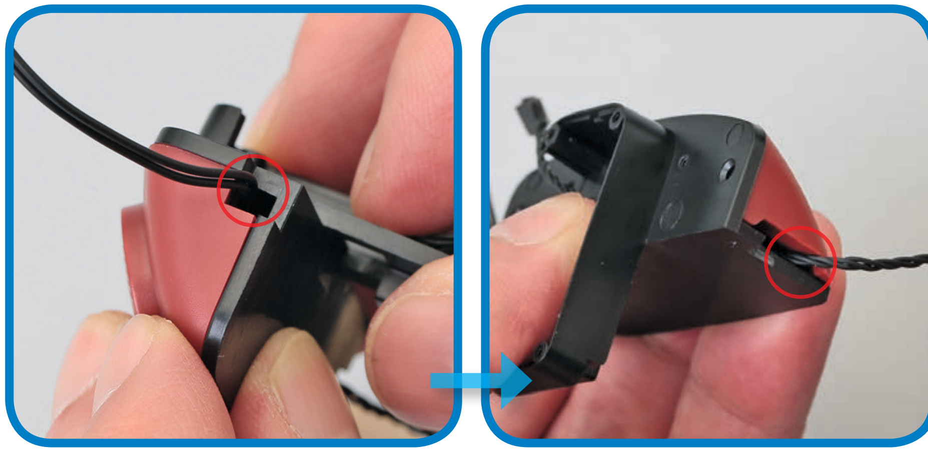
3 Secure the speaker in position (holding it carefully, as discussed), then fold its cable out and into the square cutaway in the speaker cover's side.

4 Next, take the servo cable leading from the shoulder servo on the left arm and body assembly, and feed it through the circled hole in the speaker support, from the back.



5 Pull the cable through the hole, just far enough for it to rest in position without slipping back out.

6 Now line up the speaker and its cover with the support.



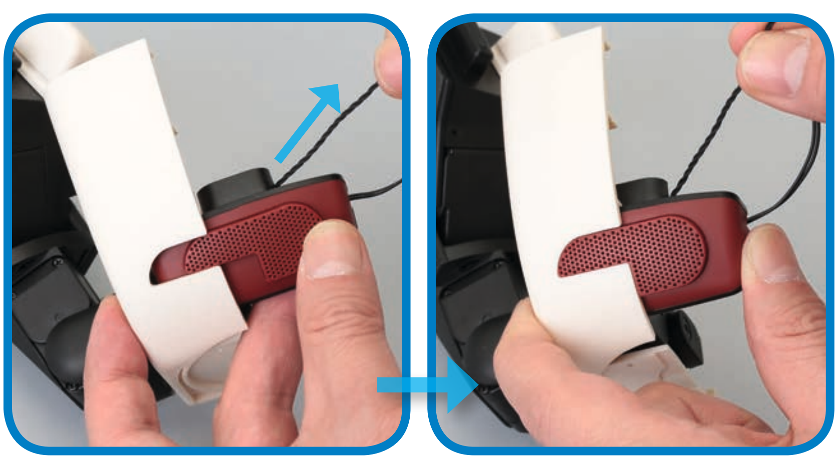
7 Press the parts together to seal in the speaker. The servo cable will sit in the cutaway on the side of the speaker cover. Again, be careful when handling the speaker.

8 Before proceeding, double-check that the speaker and servo cables are sitting in their respective cutaways without being trapped or pinched in any way.

SECURING THE SPEAKER

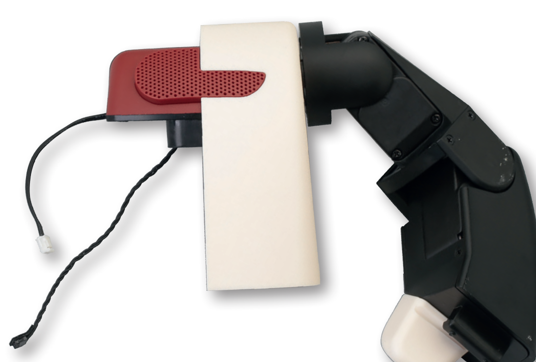
9 Use two M2 x 6mm countersunk screws to secure the speaker to its base and cover, inserting them via the two circled holes.

10 Now line up the speaker with the correspondingly shaped cutaway in the left body panel.



11 Gently pull the servo cable to remove any slack, and fit the speaker to the panel. If you struggle to fit the speaker comfortably, do not force it – instead remove it, adjust the servo cable, and try again.

Assembled speaker and left body



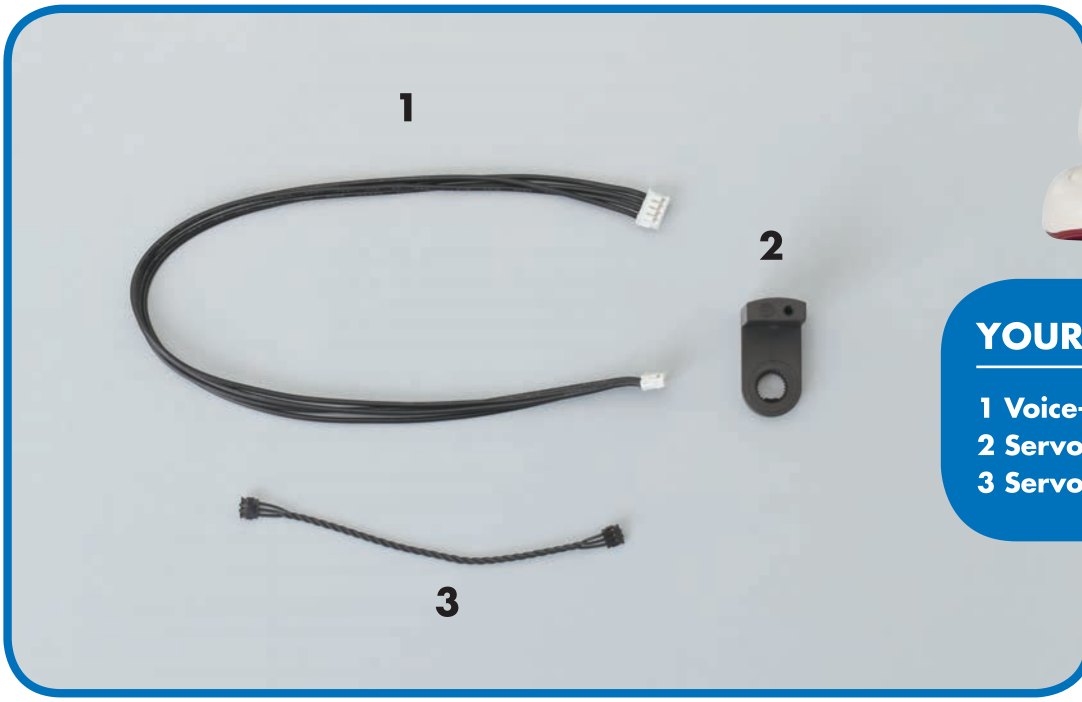
Remember, to see all of Robi's assembly stages, including this one, visit the ModelSpace web page at:

www.model-space.com

STAGE 56: MOUNT ROBI'S NECK SERVO

In this session, you will fit the neck servo prepared in Stage 54 to the neck base in Robi's scarf, wiring the scarf with the voice-recognition cable supplied here as you do so.

Robi's 'scarf' is an important part of his design, because not only does it provide a safe place by which to hold him when you pick him up, it also – along with the neck base fitted into it – provides a connection passage for the cables linking the components in Robi's head to the CPU board, located on the inside of his right body panel. In this stage, you will begin the process of wiring the scarf, and then fit the neck servo into its base.



YOUR PARTS

1 Voice-recognition cable
2 Servo horn
3 Servo cable 70mm

SAVED PARTS

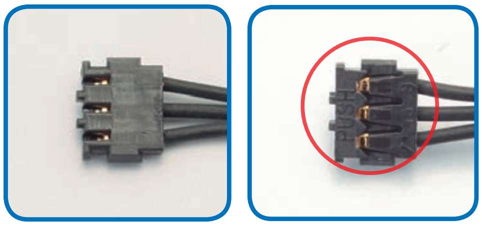
You will be using parts from the previous stages, so be sure to have these to hand before beginning.

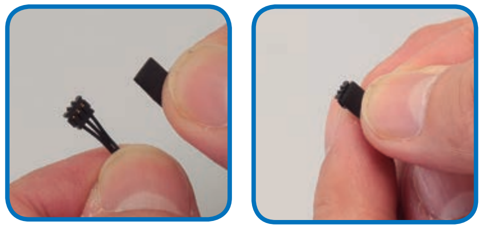
Robi's scarf and neck base (Stage 53)

Servo ID=13 (Stage 54)

Protective pads (Stage 3)

SEALING THE SERVO CABLE





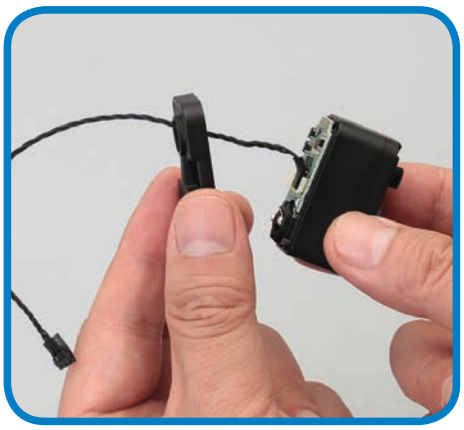
TIP!

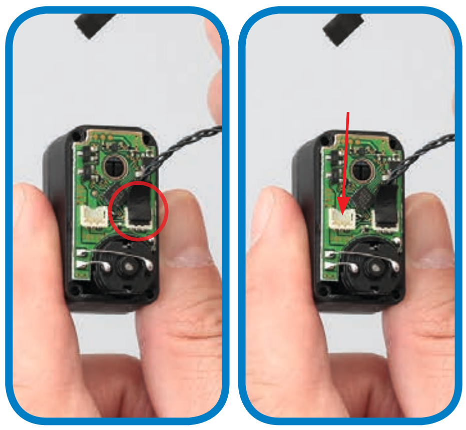
If you want to delay fitting the pads until the cable is fitted to its servo, you may do so.

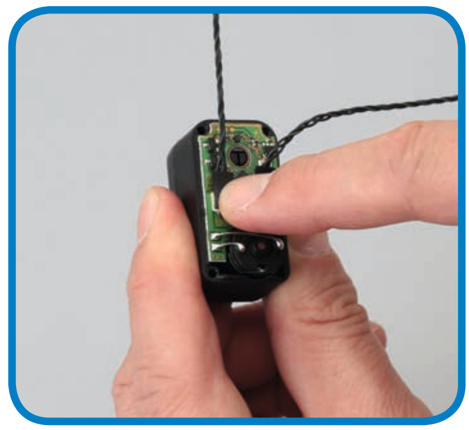
1 As you have done previously, identify the side of this stage's servo cable connector that has PUSH marked on it (circled).

2 Apply a protective pad to the connector on this side. Repeat for the other end of the cable.

WIRING THE NECK SERVO



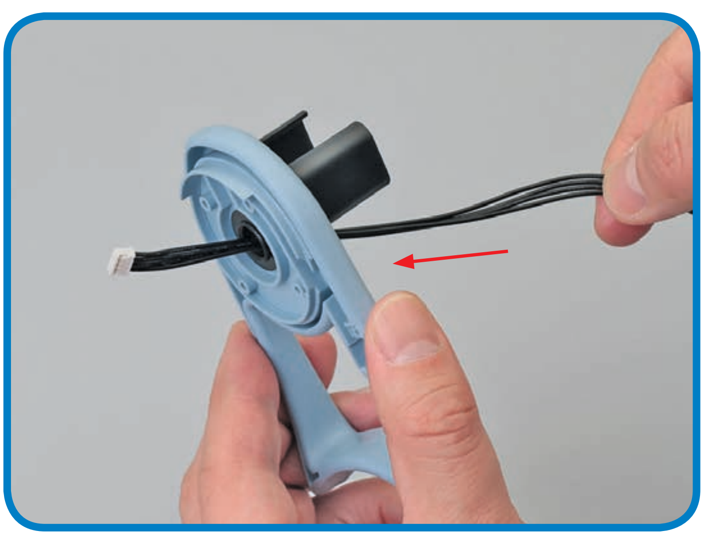




3 Remove all four screws from the neck servo ID=13, prepared in Stage 54, keeping the screws safely.

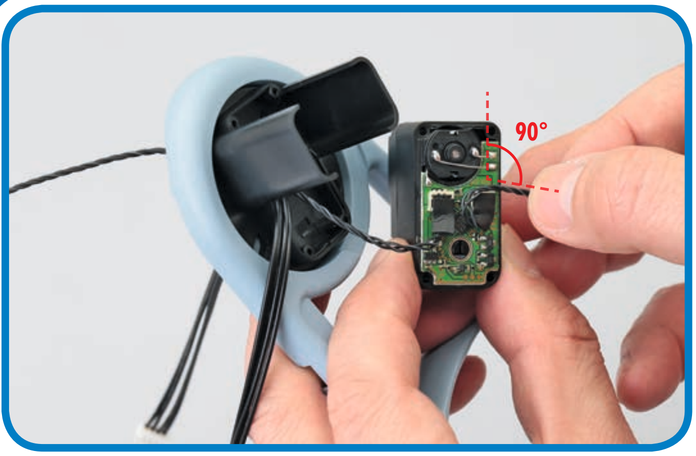
4 Check that the existing servo cable is connected securely, then fit the cable supplied in this stage into the adjacent socket.

5 Double-check that the cable is fitted securely, and if you have not already, fit its protective pad.

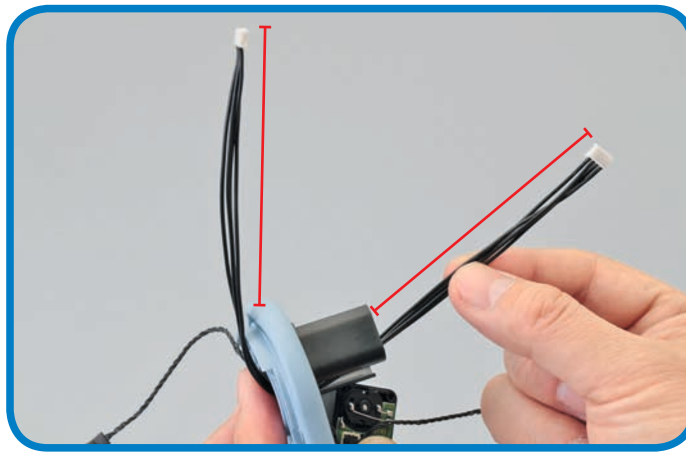


6 Take the scarf and neck base assembly from Stage 53, and feed this stage's voice-recognition (VR) cable through the hole in its centre, from the top. Stop when it's halfway through.

7 Now line up the neck servo and feed the longer of its two cables (the 135mm cable fitted in Stage 54, not the 70mm one fitted in this stage) into the same hole.



8 With the 135mm cable sitting in the hole, turn the servo over so its open face is angled towards you. Now fold the shorter cable out 90° to the side from its socket, as shown. **Make sure the connector itself remains in position.**



9 Check that the VR cable fitted in Step 6 is exactly halfway through the assembly, as indicated by the red lines in the photo above.

10 Lower the servo, with its open face down and the shaft closer to the rear of the scarf, into the neck base. Hold together the VR cable and the servo cable set at 90° in Step 8, and run both through the raised rectangular opening outlined in red above.

11 With both cables securely set within this opening, press the servo fully into the base, as shown.

12 Use two of the four screws removed from the servo in Step 3 to secure the servo via the two circled holes (keep the other screws safely for later use).

Assembled scarf and neck base

Remember, to see all of Robi's assembly stages, including this one, visit the ModelSpace web page at:

www.model-space.com

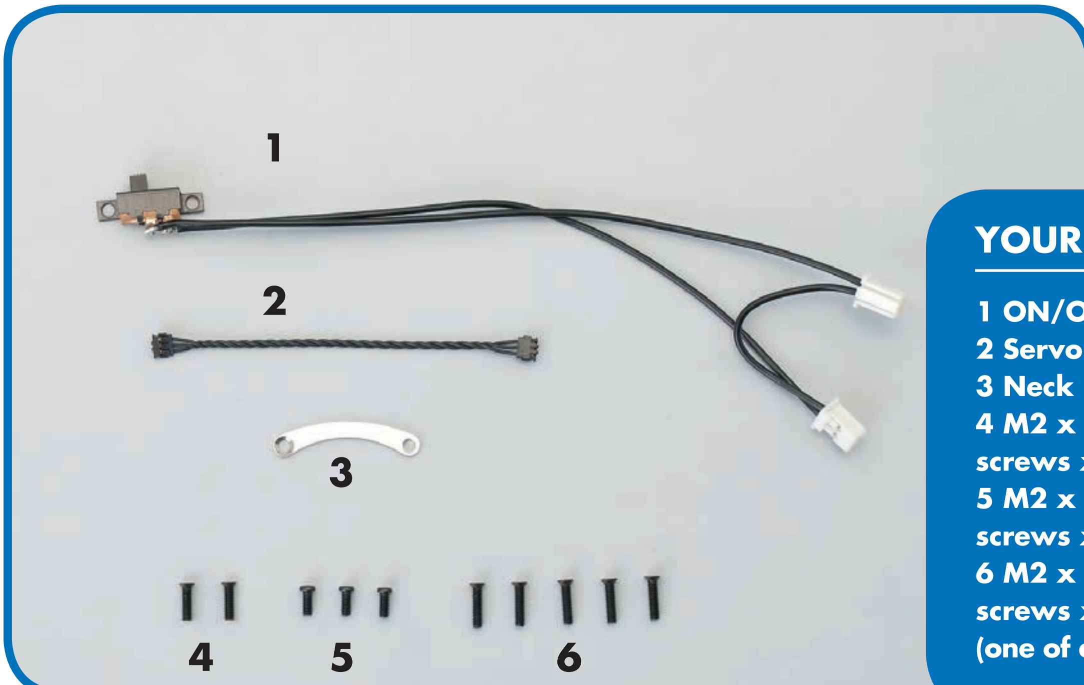
STAGE 57: WIRE ROBI'S SCARF AND BODY

In this session, you will take a significant step in the build, when you join Robi's left and right body panels for the first time. But before you do, you will need to wire them, and the scarf, carefully.

Having joined Robi's legs in the previous pack, you now follow on by joining the body, as Robi really begins to take shape. Before you can seal the parts, though, you must wire them following the instructions carefully.

This is very important, because if any of the cables get caught, pinched or arranged incorrectly, you will have to go back, disassemble the parts and start all over again! You begin by adding the ON/OFF switch.

PARTS TO BE ASSEMBLED

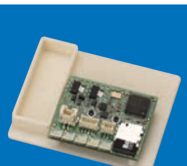


YOUR PARTS

1 ON/OFF switch
2 Servo cable 70mm
3 Neck plate
4 M2 x 6.8mm countersunk screws x 2
5 M2 x 4.5mm pan-head screws x 3
6 M2 x 8mm countersunk screws x 5
(one of each screw is a spare)

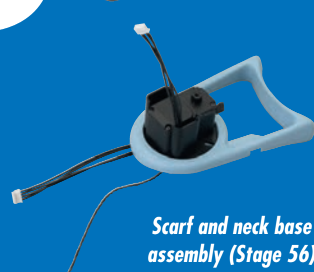
SAVED PARTS

Servo horn (Stage 56)



CPU board and test micro SD card (Stage 44)

Protective pads (Stage 3)



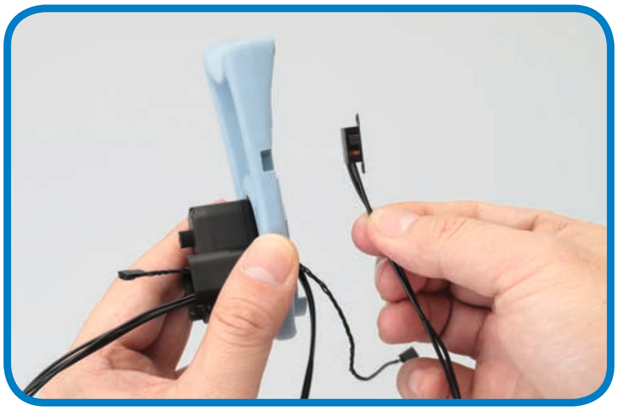
Scarf and neck base assembly (Stage 56)

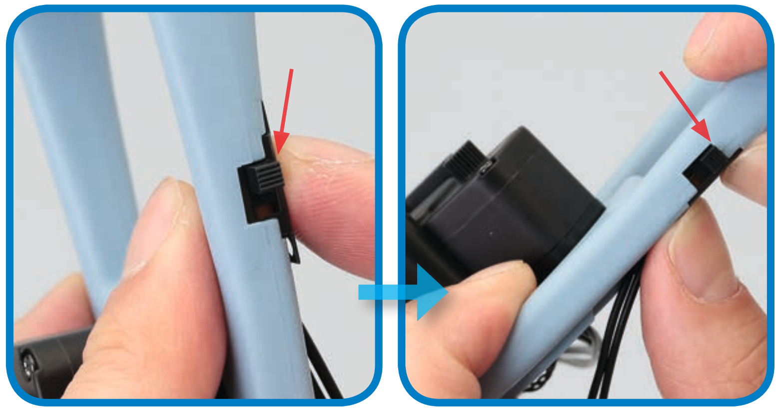
*Right body panel and arm (Stage 43)**

*Left body panel and speaker (Stage 55)**

**The arms were tested using the CPU board in Stages 43 and 44, so should work correctly. If you did not perform the movement tests then, do so before proceeding with the assembly of this stage.*

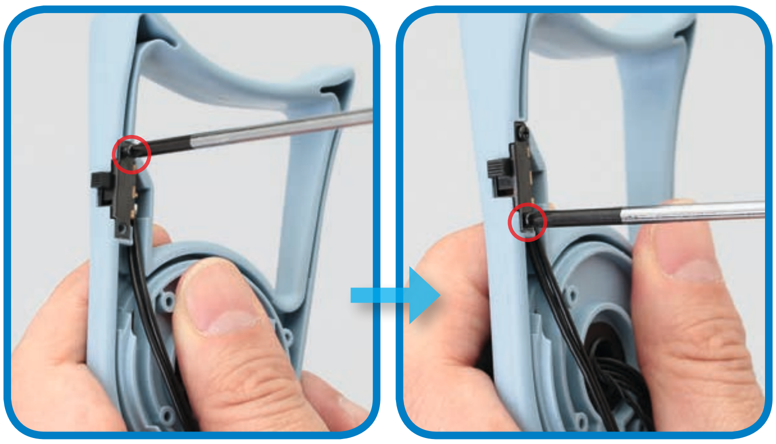
FITTING THE SWITCH

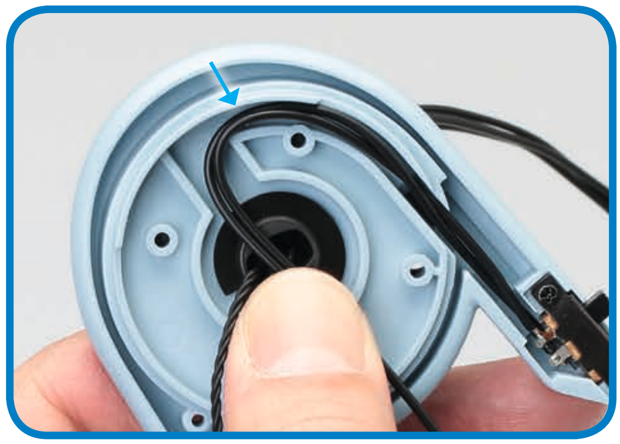




1 Line up the ON/OFF switch to the scarf, as shown.

2 Fit the switch's slider into the rectangular recess in the side of the scarf, so that its housing is flush with the part's underside.

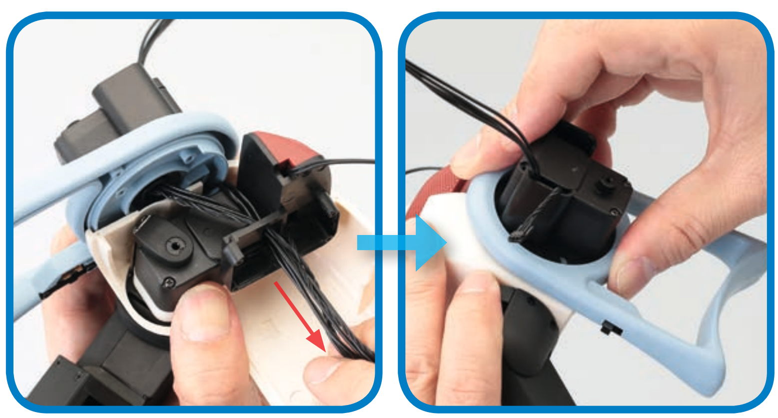




3 Use two M2 x 4.5mm pan-head screws to secure the switch via the two circled holes.

4 Run the switch's cable along the arrowed groove, then round so that it runs across the centre of the neck base. Hold this in place.

ATTACHING THE SCARF



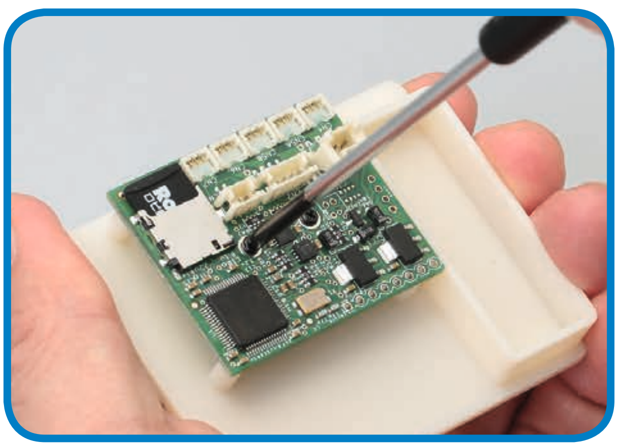
5 Hold the left body panel in one hand, and with the other, feed all of the cables running from the scarf and neck base into the circled hole on the inside of the speaker support.

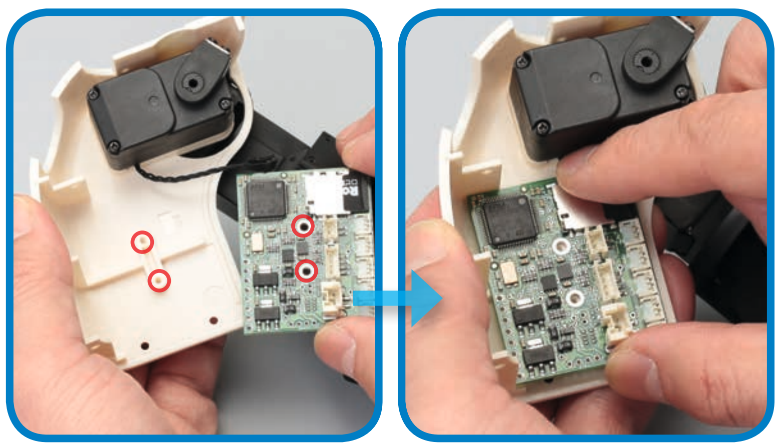
6 Lower the scarf and neck base assembly onto the top of the left body panel. Pull the cables downward gently as you do so, to prevent any build up of slack, but ensure that the switch cable remains in the groove highlighted in Step 4.

7 Rotate the neck base about 90° to the right, so that the neck servo's shaft is facing the left arm's shoulder. When the circled hole in the scarf beneath becomes visible, insert an M2 x 8mm countersunk screw and tighten it with a screwdriver. The screw should sit flush to the countersunk hole in the scarf, with the neck base still able to rotate.

8 Now turn the neck base back so the shaft faces the rear of the scarf again, and the circled hole in the scarf becomes visible. Again, insert and tighten an M2 x 8mm countersunk screw into this hole. **Do not rotate the base more than directed here, as this can twist and potentially damage the cables.**

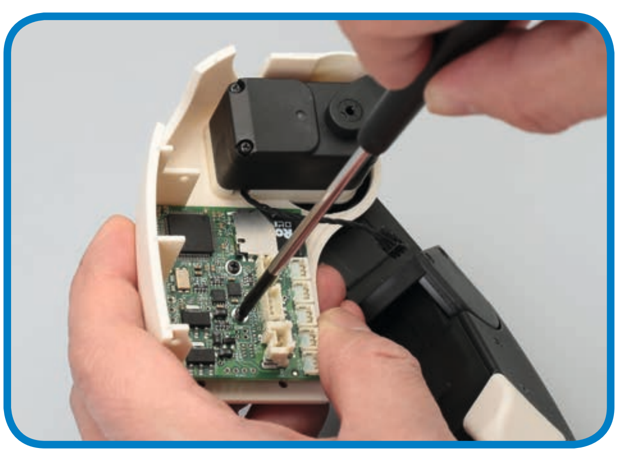
MOUNTING THE CPU BOARD

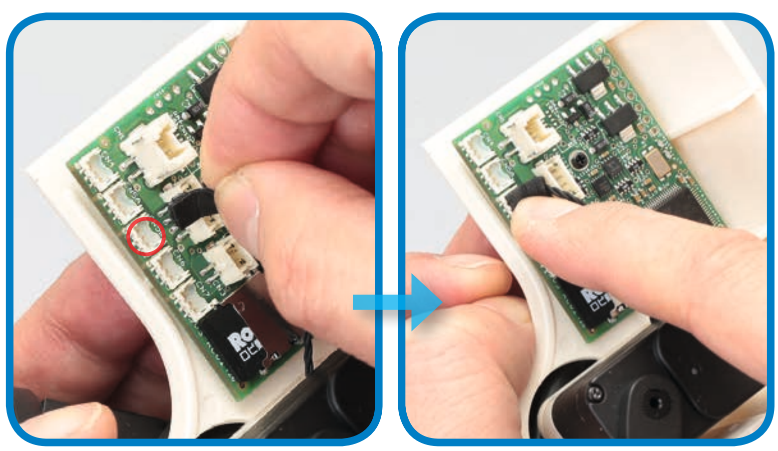




9 Take the CPU board on its support, first used in Stage 44, and remove the two screws holding the parts together. Keep the screws, but you will no longer need the support.

10 Now line up the CPU board to the inside of the right body panel, so that its two screw holes line up to those on the panel. Remember, hold the board by its edges to avoid damaging it.

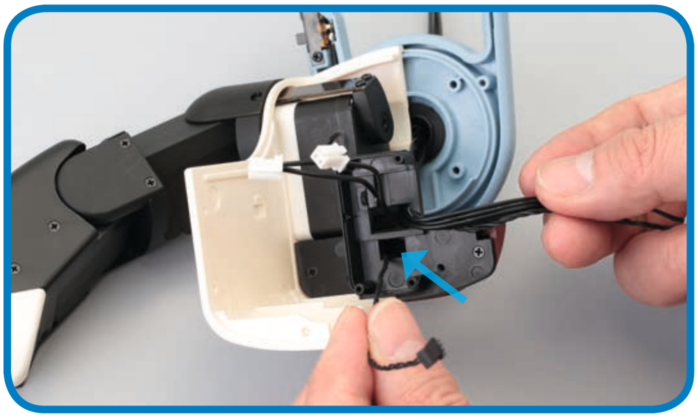




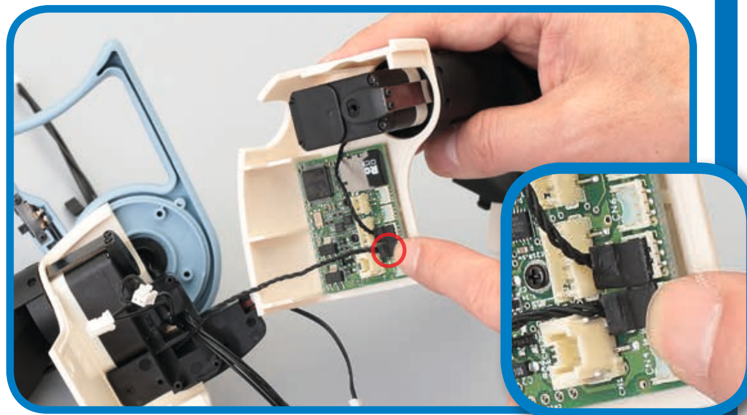
11 Use the screws removed in Step 9 to secure the CPU board.

12 Connect the servo cable running from the right arm's shoulder servo to the right arm socket on the CPU board, circled.

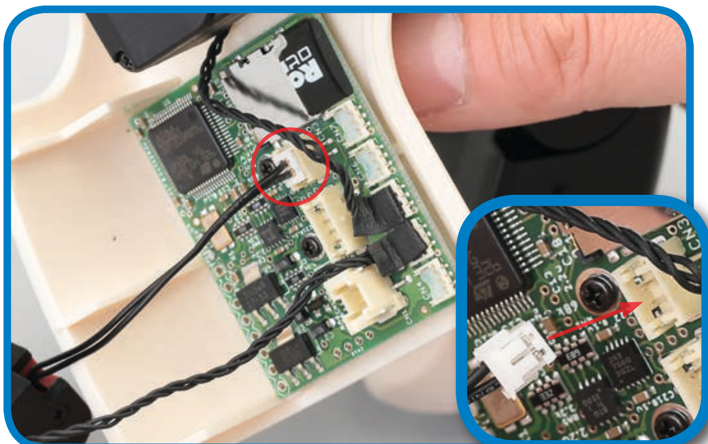
JOINING THE BODY PANELS



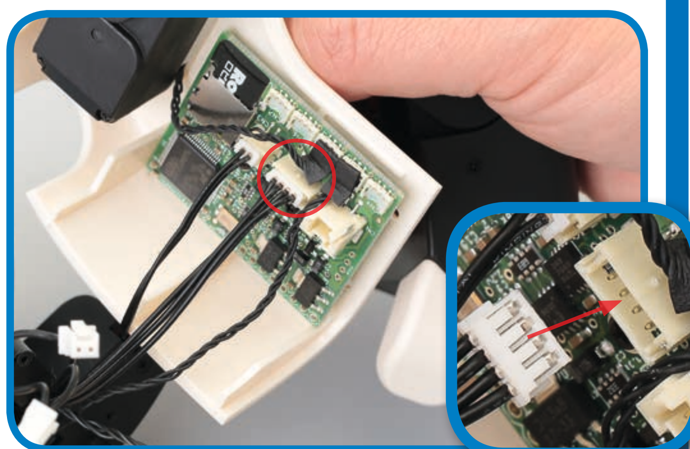
13 Take the left body panel and locate the servo cable leading from the left shoulder servo, which is fitted through the hole in the speaker support (arrowed). Separate this from the other cables.



14 Lay the left body panel out on your work surface, positioned as shown, then line up the right body panel to it. Fit the servo cable separated in the previous step to the left arm socket, next to that of the right arm (inset).

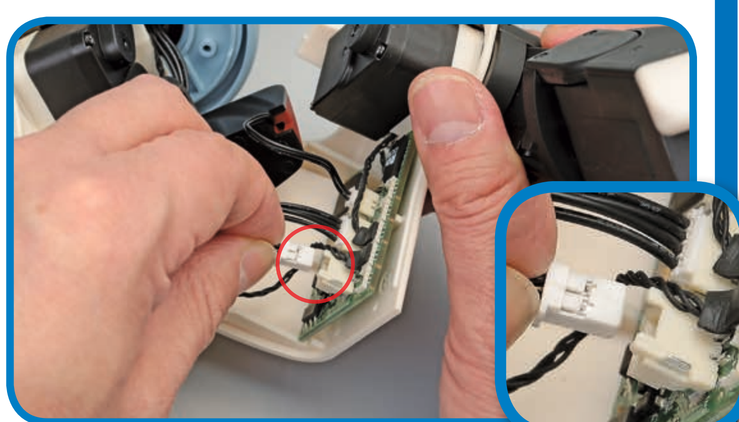


15 Now locate the two-strand cable running from the speaker, and fit its connector into the circled socket (the speaker socket, as indicated on the CPU board diagram in Pack 12). Be careful not to force the connector, and to set it perfectly straight when inserting it, to protect the delicate metal pins inside the socket's housing.

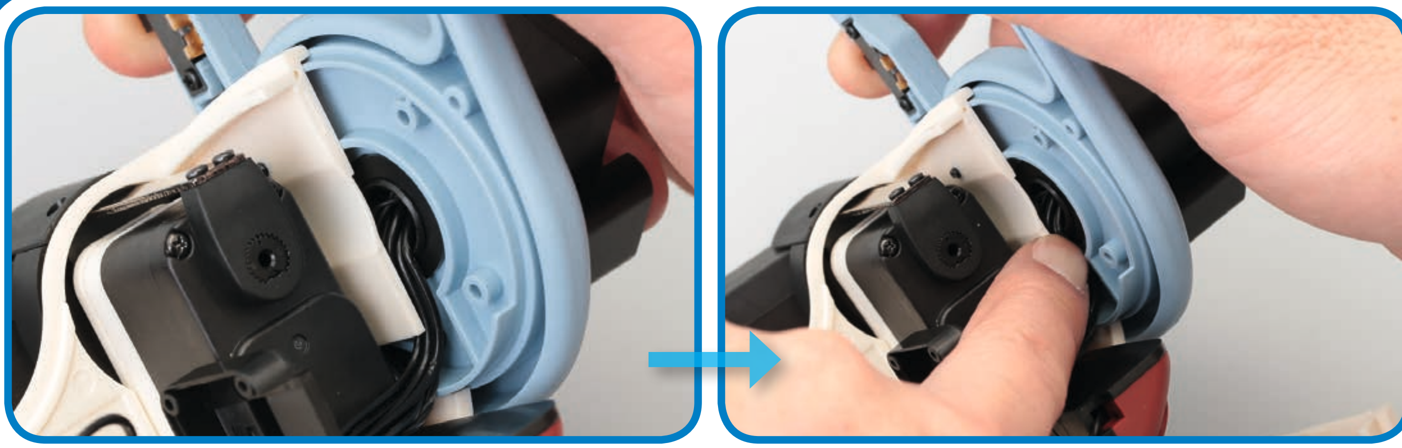


16 Next, take the four-strand VR cable and fit its connector into the middle socket. Again, be careful when inserting the connector.

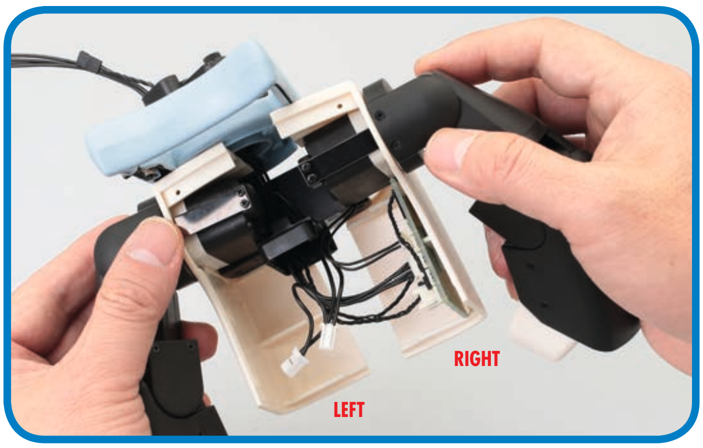
17 Next, find the ON/OFF switch's cable, which has two white connectors at its end.



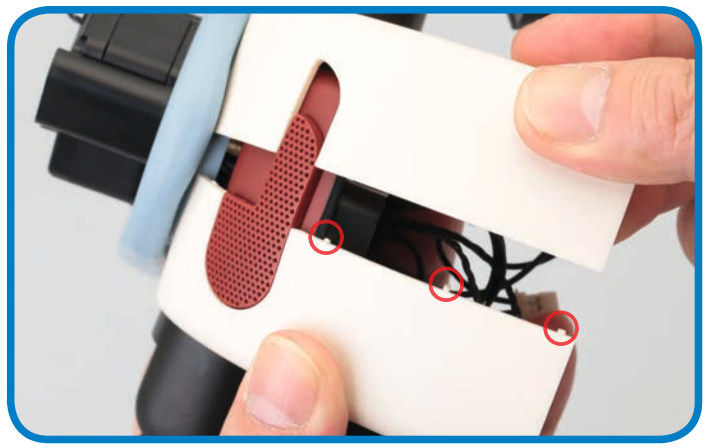
18 Insert one of the connectors into the circled socket, taking care not to force the part as you do so. It does not matter which one you use.



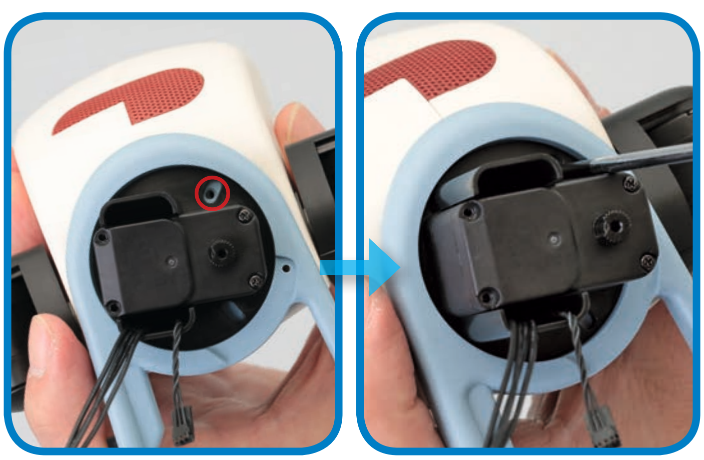
19 You are now ready to join the body panels, so begin by routing all the cables leading into the scarf/neck base around the cream-coloured plastic of the left body panel, as shown. This will prevent them from being trapped as you join the left and right panels.



20 Bring the left and right panels together.



21 Join the panels so that the three circled locating pins secure the parts correctly. If you feel any resistance, this is probably due to cables bunching up inside, so simply re-route these until it fits easily.

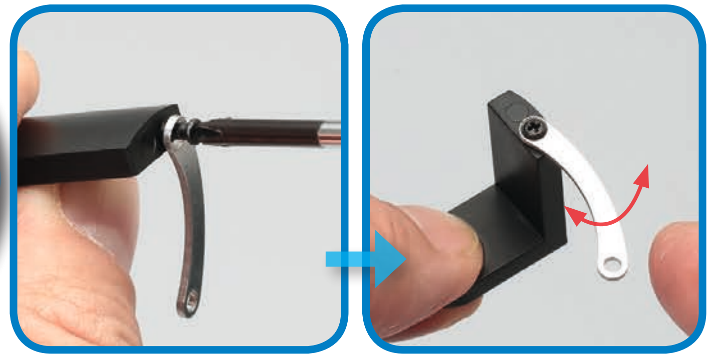


22 Holding the left and right body panels together, turn the neck base to the left so the circled hole is accessible. Then insert and tighten an M2 x 8mm countersunk screw into the hole. The screw should sit beneath the neck base, allowing it to rotate freely.

23 Turn the neck base a little further round, until the circled hole (on the other side of the neck servo to the one used in the previous step) lines up with the one in the scarf beneath. Fit another M2 x 8mm countersunk screw to complete the process of joining the body panels.

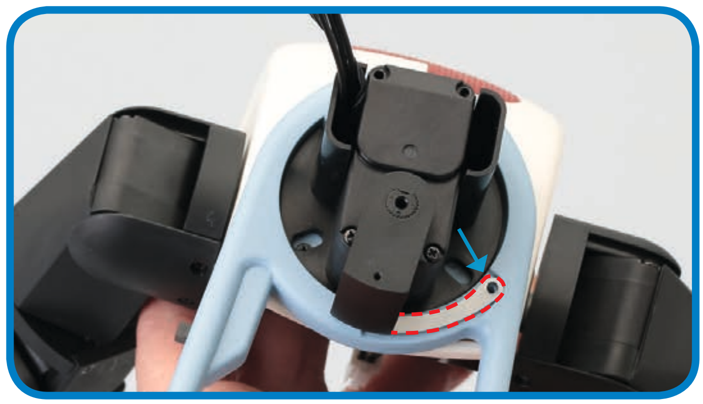
ADDING THE NECK PLATE

24 Next, line up the neck plate with the end of the servo horn that has a countersunk hole in it. Make sure that the hole's countersunk side is facing up, ready to accommodate the screw head.

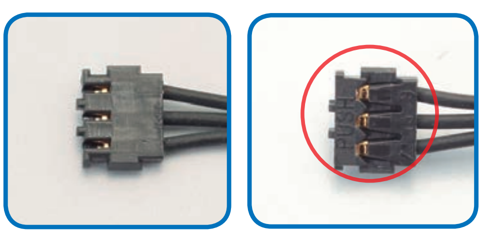


25 Insert and tighten an M2 x 6.8mm countersunk screw, but don't tighten it fully – the plate must be able to swivel from side to side, as shown above right.

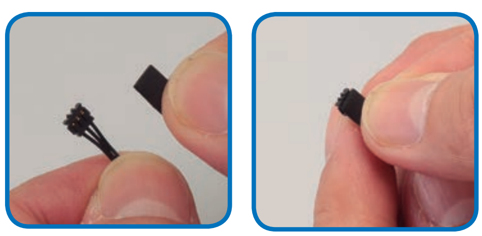
26 Line up the servo horn's D-cut hole to the shaft of the neck servo, with the neck plate level with the scarf. Join the parts, as ever taking care to match the D-cuts perfectly.



27 Seat the neck plate in the correspondingly shaped recess in the scarf, outlined above. The hole at its end should exactly match the hole in the scarf.



28 As you have done previously, identify the side of this stage's servo cable connector that has PUSH marked on it (circled).



29 Apply a protective pad to the connector on this side. Repeat for the other end of the cable.

If you want to delay fitting the pads until the cable is used in the build, you may do so.

Remember, to see all of Robi's assembly stages, including this one, visit the ModelSpace web page at:

www.model-space.com

Assembled body

STAGE 58: PREPARE AND FIT ROBI'S BACK SERVO

In this session, you will prepare then fit servo ID=12, the unit that will turn Robi's body from side to side.

This session sees you programming a servo again, in this case, the back servo that will rotate Robi's body from side to side, ID=12. Once you have programmed the back servo, you will fit it into the specially shaped frame in the speaker support, set on the inside of Robi's body panels. Take care during this stage of the build, because you will be working close to a number of delicate and carefully positioned cables.

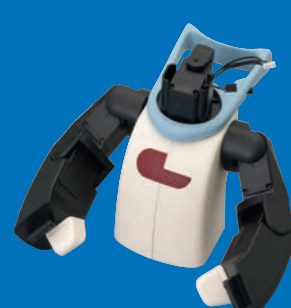
YOUR PARTS

1 Servo motor (ID=12)

SAVED PARTS

You will be using parts from the previous stages, so be sure to have these to hand before beginning.

Servo cable (Stage 57)



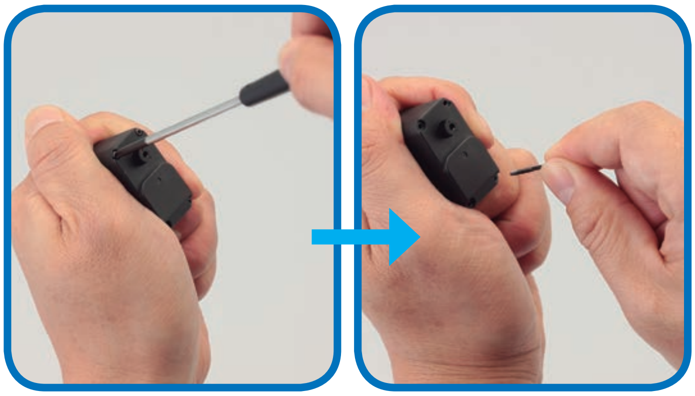
Robi's body (Stage 57)

Neck stand (Stage 8)

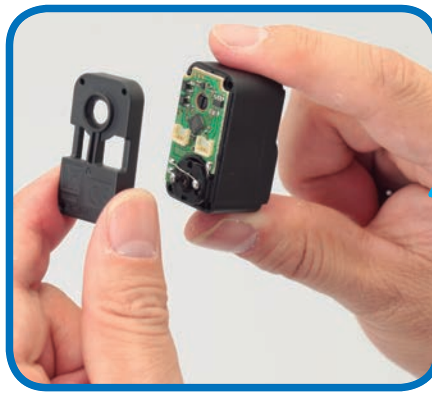
WIRE THE SERVO

TIP!

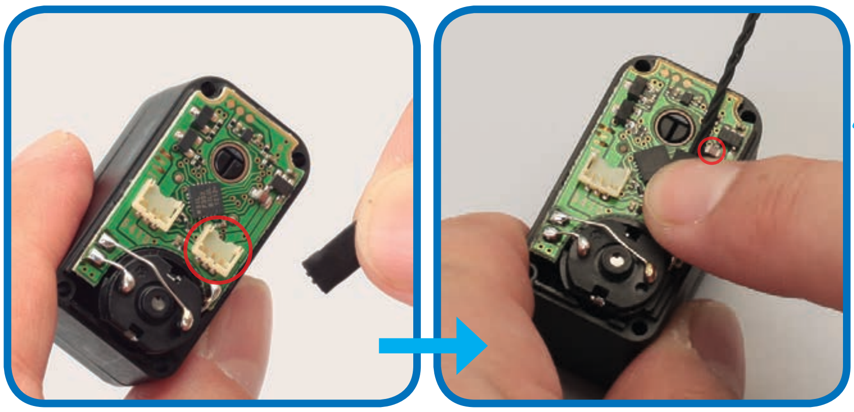
As ever, it is very important that the electronic parts exposed by removing the servo's cover are not touched in any way. Make sure you hold the open servo by its sides, and if you put it down on your work surface during the assembly, do so with the open side facing upwards.



1 Take out the four screws holding the servo's removable cover in place, and keep these safely to one side.



2 Gently lift the cover away, taking care not to force it or damage the circuit board beneath it.



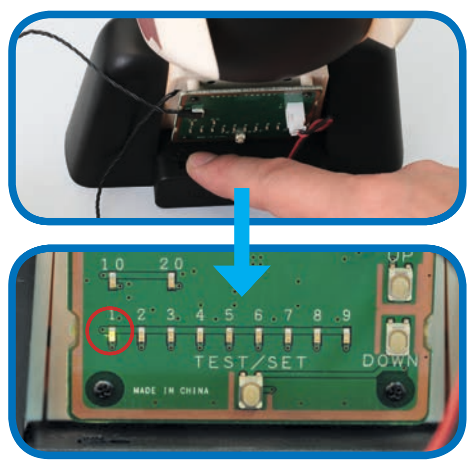
3 Hold the open servo with the circuit board facing upwards. Fetch the servo cable fitted with protective pads in the previous stage, and press one of its connectors into the socket circled in red, as you have done before. It does not matter which end of the servo cable you use.

If you chose not to fit the protective pad in the previous stage, do it now. Also take extra care not to touch the circled silver component on the servo's open board.

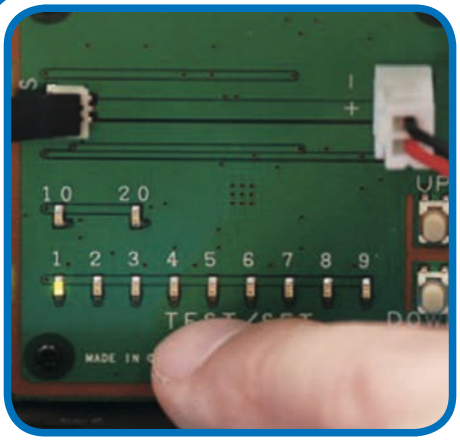
4 Replace the servo cover, running the cable through its central hole as you do.

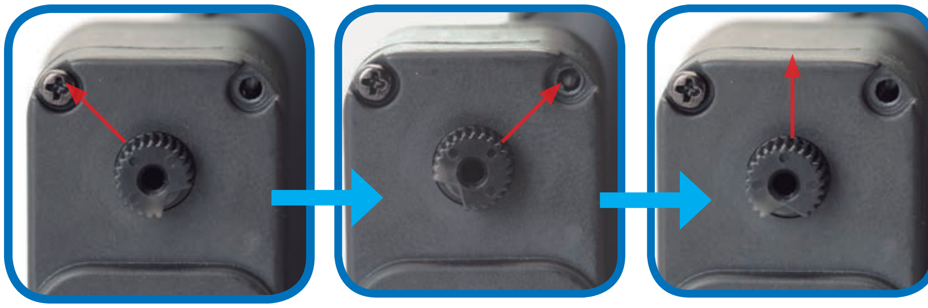
5 Fasten the cover back in place with two screws, but tighten them only lightly because you will be removing them again later.

6 Make sure the test board is switched OFF, then connect the servo's cable to it.



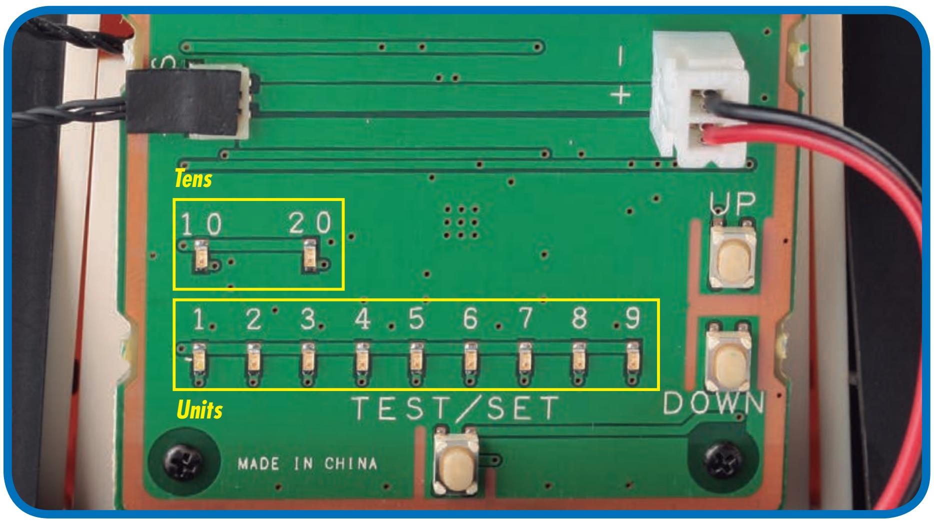
7 Turn the board ON. The LEDs should flash, then leave the 1 LED lit alone.



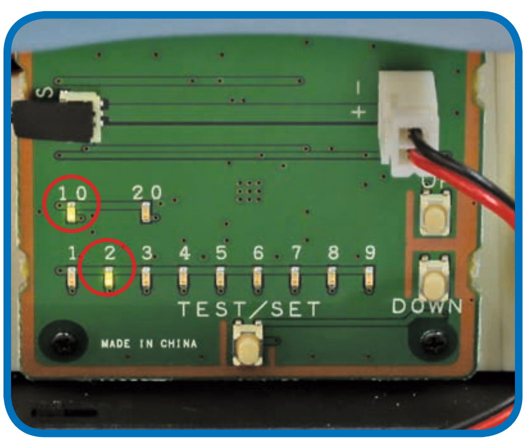


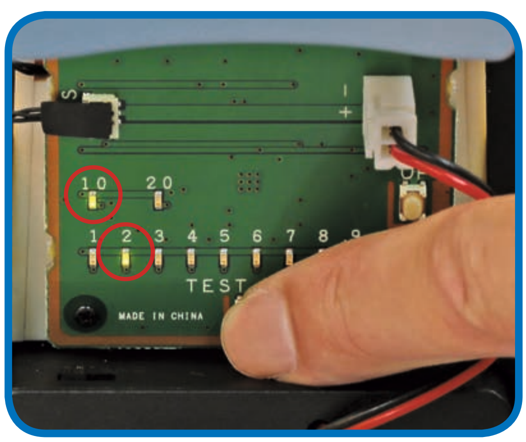
8 Test the servo by pressing the TEST/SET switch: the servo's shaft should rotate 45° to the left, stop, then move back across to 45° to the right. The next step is to set the servo's ID number, so keep the power on. If the test does not run, or if the LEDs flash rapidly, refer to the HELP box in Stage 8.

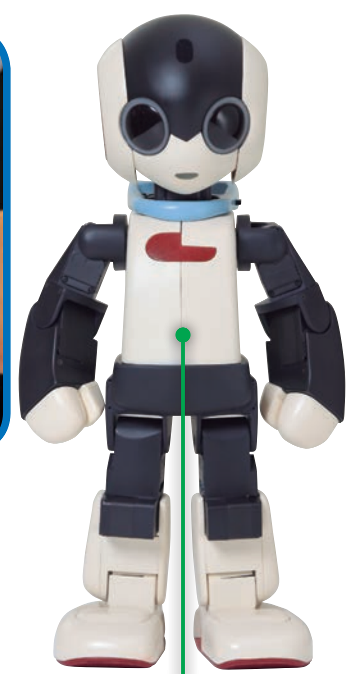
SETTING THE ID NUMBER



9 Next, you will set the ID number of the servo using the test board, as you have done before. Each time you press the UP switch, the ID number increases by one, and the result is displayed by the two banks of LED lights. The top row reads in tens and the bottom in units. For example, to set the ID to 12, as you will do in this stage, you press UP 11 times. This will light up the 10 and 2 LEDs. (If you overshoot, press the DOWN switch to take it down one step at a time.) When the ID number is correct, set it by pressing and holding the TEST/SET switch.





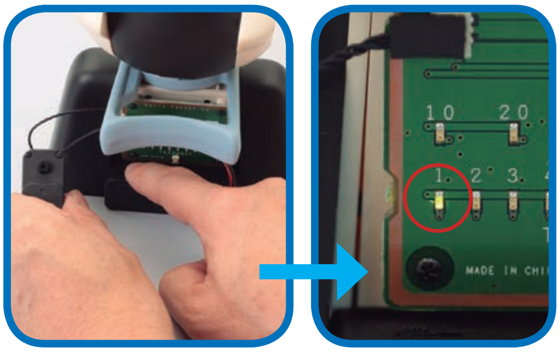


10 To set the ID number of Robi's back servo to 12, press the UP button 11 times. Make sure that only the 10 and 2 LEDs are lit, using the UP and DOWN buttons to correct the number if you overshoot when setting. Confirm the setting by pressing and holding the TEST/SET button for a few seconds. The 10 and 2 LEDs should flash a few times, then remain lit.

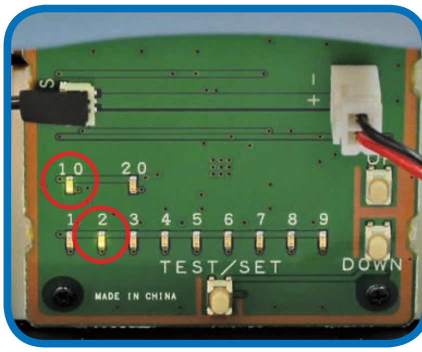
As you will have seen over the previous packs, the process of disassembling and resetting a servo's ID number can be time consuming – so take great care to set it correctly now, and be sure to test it before continuing the build! (Instructions given on the next page).

ID=12

CHECKING THE ID NUMBER



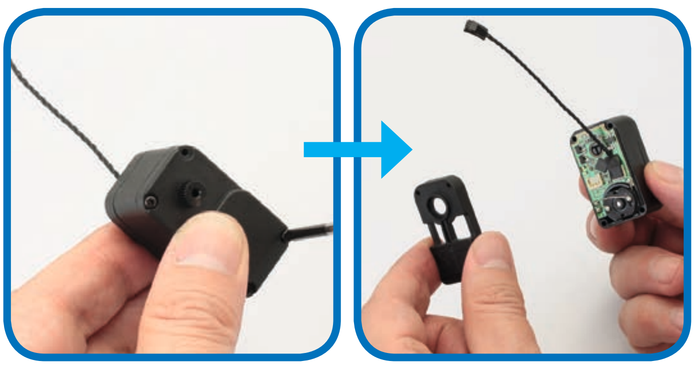
11 It is vital that Robi's servos are set correctly, so it is always a good idea to check that you have set the intended ID number afterwards. To do this, leave the servo connected, then switch the power OFF, then back ON. Only the 1 LED should light up.



It is normal for the 1 LED to light up when you switch on. It will only change to give you the ID number of the servo when you press TEST/SET. If you made a mistake, or set the wrong ID number, you can reset it by going through the setting procedure again.

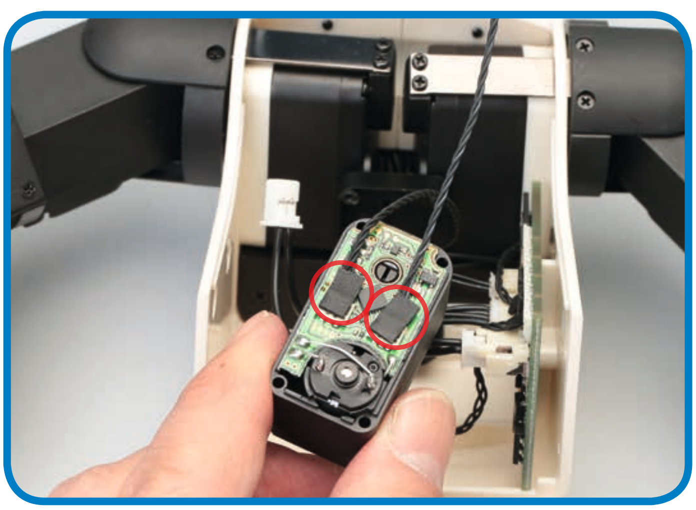
12 To check the ID number, press TEST/SET briefly. Now the 10 and 2 LEDs should light up and the servo shaft should repeat the rotations seen in the test. Note: if you find that a servo has the wrong number assigned to it, go back and repeat the setting process in full.

FITTING THE BACK SERVO

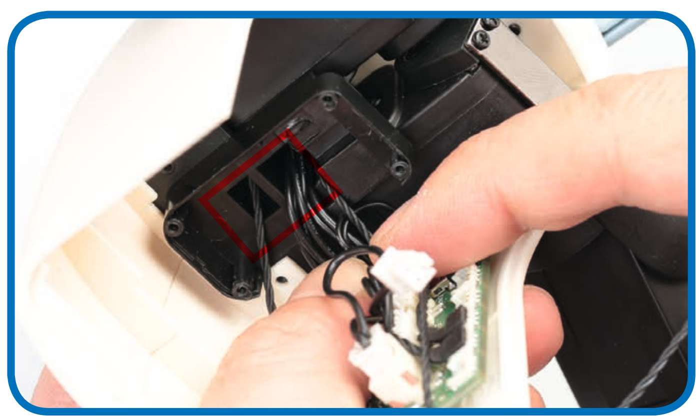


13 Remove the cover fitted in Step 5. Keep the screws, but you no longer need the cover.

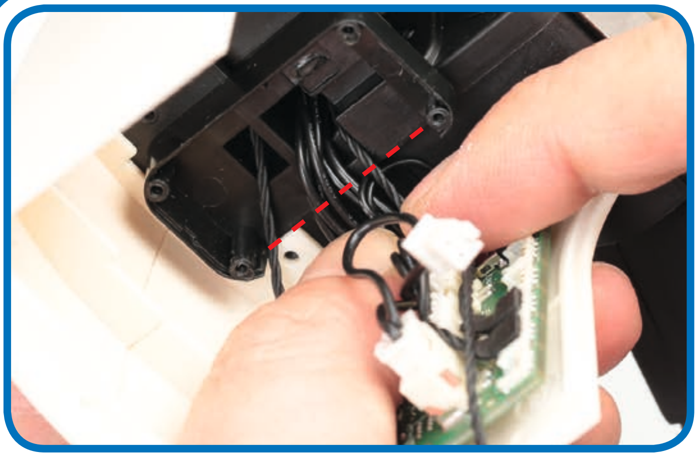
14 Line up the back servo to the open back of the body assembly, then fit the free end of the neck servo's cable into the free socket on the back servo.



15 Check that both connectors are securely attached to their respective sockets.

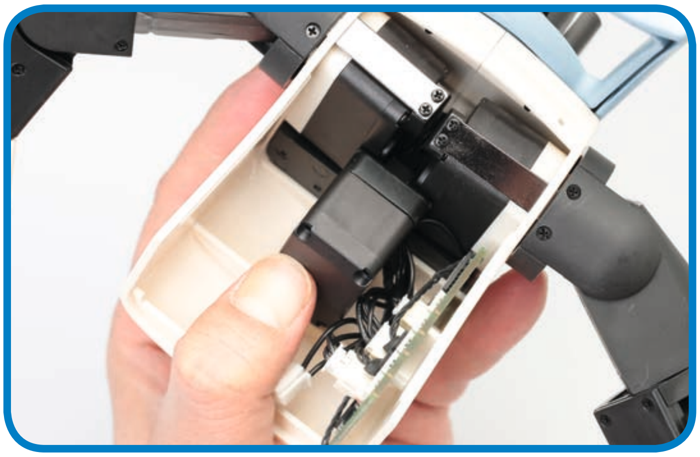


16 Making sure that the back servo's connectors stay in position, reach into the body assembly with your fingers and thumb. Now fold the set of cables running out from holes in the speaker base (outlined in red) 90° towards the CPU board.

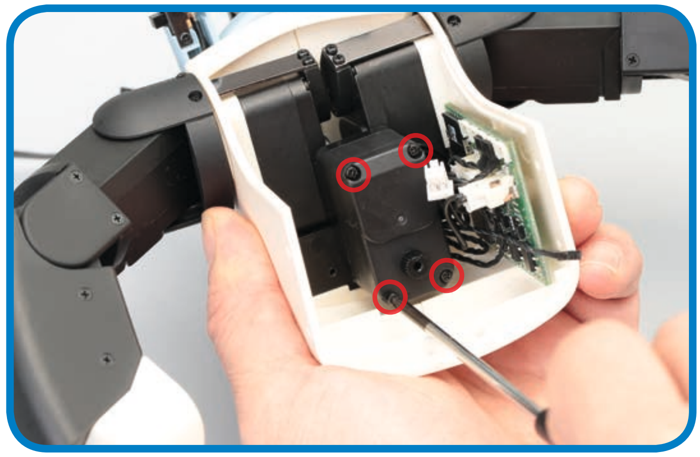


17 Arrange the cables so that they all sit neatly along the flat, unwalled side of the moulded servo frame (dotted red line). Apply a little pressure on the cables so that they 'set' flush to this edge, as shown.

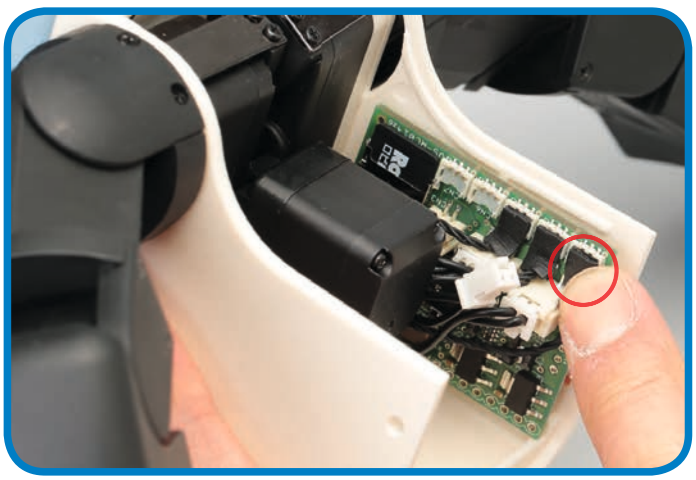
18 Now, very gently fit the back servo to the frame, with its shaft closest to Robi's chest. Do this very slowly and carefully, so that the cables are not crushed and the servo connectors are not disturbed.



19 Once you are absolutely sure that none of the cables is trapped, crushed or dislodged, press the servo firmly into its frame. If necessary, lift it away, reset the cables and repeat until the parts are neat and secure.



20 Now use the four screws removed in this stage to secure the servo.

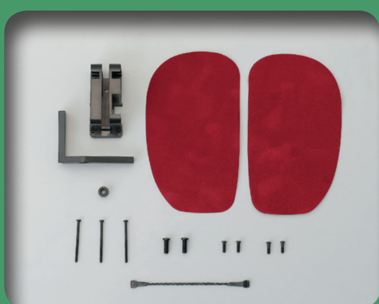


21 You can now fit the free end of the back servo's cable into the head/neck/back socket on the CPU board, farthest from the micro SD card slot.

Assembled body

COMING IN PACK 16

Your next **FOUR** complete Stages, as Robi starts to come to life!

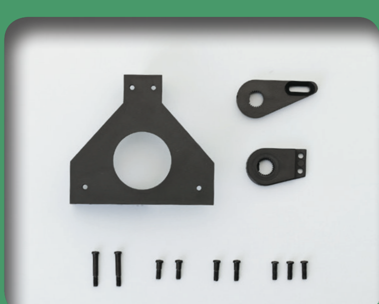


Stage 59
Prepare Robi's first head servo

PARTS PROVIDED
Head servo frame and foot pads.

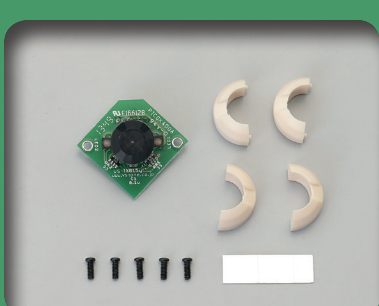
Stage 60
Prepare Robi's second head servo

PARTS PROVIDED
The second of Robi's head servos.



Stage 61
Continue building up the components inside Robi's head

PARTS PROVIDED
More internal head components.



Stage 62
Fit the first of Robi's infrared sensors

PARTS PROVIDED
Infrared motion sensor and frames.